

Star edge colourings of generalized Petersen graphs and certified reductions towards the six-colour conjecture

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[Code and formal proofs](#)

Abstract

The latest construction gives five-star edge colourings of $GP(n,k)$ for $1 \leq k \leq 15$ and $n \geq 2k+1$, except $GP(3,1)$, alongside other infinite families, small cubic multigraph classifications and conditional reduction theorems. The general six-colour conjecture remains open.

$\chi'_s(GP(n,k)) \leq 5$ for $1 \leq k \leq 15$, $n \geq 2k+1$, $(n,k) \neq (3,1)$; additionally selected infinite-family colourings, finite six-colour cases and explicitly conditional reduction theorems.

Definitions and principal unconditional scope

A star edge colouring is a proper edge colouring with no two-coloured path or cycle of four edges. Parallel edges in a loopless multigraph are distinct and adjacent. The Dvořák-Mohar-Šámal conjecture asks whether every subcubic multigraph has star chromatic index at most six; the located general published bound is seven.

The strongest current generalized-Petersen endpoint states that $GP(n,k)$ has a five-star edge colouring for $1 \leq k \leq 15$ and $n \geq 2k+1$, except $(n,k) = (3,1)$. Here $GP(n,k)$ has an outer n -cycle, n spokes and inner edges of step k . The condition $n > 2k$ makes the construction an ordinary simple cubic graph. This theorem includes parameter regions not covered by the earlier even-order or gcd-based results.

Additional explicit endpoints give colourings of flower and Goldberg snarks, Möbius and inflation families, and a matching-colour-class six-colour construction for $GP(n,2)$. Known parameter subfamilies remain background to these constructions.

Periodic blocks, seams and local-window lifting

The $GP(n,k)$ construction colours repeated blocks and inserts a finite seam determined by the residue of n modulo a selected period. Each k has an explicit ordinary graph colouring recipe, small exceptional tables and representative windows. For example, the $k=4$ development repeats a period-five block, handles $9 \leq n < 19$ directly and compares later windows with representatives $25 + (n \bmod 5)$.

A forbidden four-edge path or four-cycle touches only a bounded range of blocks. The `BlockStar.star_of_windows` theorem embeds such a candidate into a window of length $3D+1$, where D bounds the relevant edge jump. A finite representative-window check therefore rules out every forbidden pattern in all large n . This is an infinite proof supported by finite local certificates, rather than an inference that a finite experimental table probably continues.

The independent ordinary audit regenerated 1,366 GP instances across $k=1,\dots,15$, including all small orders through 100 and large examples near 1000, and checked proper colouring and every relevant four-edge walk. Fifty-one flower-snark examples also passed. All GP JSON tables matched the actual Lean literals. These 1,417 tests support semantic fidelity and the finite data, while the formal local-window theorem supplies the intended unbounded conclusion.

Small multigraphs and the reduction programme

The unconditional finite endpoints include six-star edge colourability of all bridgeless cubic loopless multigraphs with at most 14 vertices, and of all subcubic multigraphs with at most seven vertices. They are not interchangeable: lifting a cubic census result to arbitrary subcubic graphs can increase the number of vertices, so the 14-vertex threshold cannot simply be copied.

The reduction library organizes a general loopless-subcubic theorem into cubic and leaf statements. Its basic equivalence says the full conjecture follows from six-colourability of bridgeless cubic graphs together with the six-colourability of their appropriate leaf attachments. Components, bridges and a degree-two core are handled by restriction and gluing lemmas. Two-edge-cut decomposition and digon insertion then concentrate the unresolved cases on more structured cubic hosts.

The library also introduces perfect-matchings-as-colour-classes, far-exchange sets and pole-dominance data. An MC colouring has one prescribed perfect matching as its sixth colour class. EX1-goodness asks for compatible choices preserving an edge's membership status; pole dominance encodes the boundary colour compatibility needed to glue cubic pieces. Theorems prove implications from precisely stated remaining hypotheses. They do not prove those hypotheses globally.

Prior work and the actual new regions

The Omoomi-Vahid Dastjerdi paper proves all $\gcd(n,k)\geq 3$ and selected gcd-two cases, and discusses earlier Zhu-Shao even/gcd-one classes. It does not establish all odd $n\geq 7$ for $k=2$. The explicit odd $k=2$ construction is a candidate new infinite region of GPAll; its proof replaces earlier incomplete $k=2$ and $k=3$ arguments.

Known covers of Petersen and K_4 , selected even odd- k classes and earlier periodic families provide prior constructions. Historical novelty of the remaining parameter regions and reduction capabilities requires an exact theorem-by-theorem comparison; introducing a definition does not by itself establish a new obstruction-removal theorem.

An optional strong-five-colour hypothesis may be false in light of the Deng-Liu-Feng 2025 infinite cubic family requiring six colours. Its exact graph assumptions require full-text comparison. An implication from that hypothesis can remain logically valid, even if the hypothesis is false; it would then not be a route proving the conjecture. This concern does not invalidate the explicit GP and flower endpoints.

Proof scope and open premises

The formal statements verify the finite constructions, infinite-family constructions and conditional reduction implications as typed. The five infinite structural premises are not proved globally, and the general DMS conjecture remains open. A selected endpoint audit does not imply that every theorem in the source portfolio has the same axiom classification.

The supplementary mathematical exposition gives the generalized-Petersen periods, seams and exceptions for $1 \leq k \leq 15$, with 519 finite instances and 20,612 representative windows checked separately. It also explains the I/II reduction, two-/three-cut gluing, prescribed matching-bit induction and universal host/marking census bridge. The base12, simple14 and b14d finite assemblies replace the earlier partial census data. The five infinite structural premises remain open.

The explicit construction scope excludes all $k \geq 16$ and does not settle unrestricted cubic or subcubic star six-colourability.

Formal verification

A qualified body-identical import-only composite route checks all 228 scientific source bodies through 221 identity-checked prior outputs and seven new source checks. Four import headers differ from the frozen originals; statements, proofs, options and comments are unchanged. The original-source run remains partial at 119/228. Its original organizer audit produced 120/123 outputs and failed because Star6Equiv was not imported; a separate supplement adds that import and checks all 123 unchanged requests. Among these selected declarations, 120 use standard kernel axioms and three are axiom-free; none uses admitted or unknown axioms. MGraph.k4subdiv_star6 in the imported StarCert finite certificate uses native_decide outside the selected set, so the entire portfolio is not asserted standard-only. The five open structural premises and the general conjecture remain open. Exact source/dependency identities and the separate audit records are supplied in the companion repository.

gp_star5

```
theorem gp_star5 (k n : Nat) (hk : 1 ≤ k) (hK : k ≤ 15) (hn : 2 * k + 1 ≤ n) (hex : ¬ (n = 3 ∧ k = 1)) :
  ∃ c : Fin (gp n k).m → Fin 5, (gp n k).Star 5 c
```

Source: entries/dms-star6/artifact/lean/pack4/src/GPAll.lean:25.

Source SHA-256: e831e584837dc1e0a2e4ef26bf7875bebd3a18ad54deff326b3c68d1af970893

gp_family

```
theorem gp_family (k n : Nat) (hk : 1 ≤ k) (hK : k ≤ 15) (hn : 2 * k + 1 ≤ n) (hex : ¬ (n = 3 ∧ k = 1)) :
  StarFam (gp n k) 5
```

Source: entries/dms-star6/artifact/lean/pack4/src/GPAll.lean:45.

Source SHA-256: e831e584837dc1e0a2e4ef26bf7875bebd3a18ad54deff326b3c68d1af970893

Sources and attribution: 1. arXiv source: 2410.15024v1; 2. arXiv source: 1011.3376; 3. DOI: 10.7151/dmgt.2195; 4. link.springer.com/s40840-025-01875-9; 5. [alejandrozu/ulam-opdp-difficulty-atlas](https://alejandrozu.github.io/ulam-opdp-difficulty-atlas/); 6. [alejandrozu/openmath-2026-judging/E08-DMS.txt](https://alejandrozu.github.io/openmath-2026-judging/E08-DMS.txt); Proof source repository.

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